



Locating roots by a change of sign

- 1 Show that $f(x) = x^3 - 4x - 2$ has a root between $x = 2$ and $x = 3$.
 $f(2) = 8 - 8 - 2 = -2$. $f(3) = 27 - 12 - 2 = 13$. Since $f(2) < 0$ and $f(3) > 0$, the sign change confirms a root lies in $(2, 3)$.
 - 2 Show that $f(x) = x^3 - x - 1$ has a root between $x = 1$ and $x = 2$.
 $f(1) = 1 - 1 - 1 = -1$. $f(2) = 8 - 2 - 1 = 5$. Since $f(1) < 0$ and $f(2) > 0$, the sign change confirms a root lies in $(1, 2)$.
 - 3 Show that $f(x) = \ln x - 2 + x$ has a root between $x = 1$ and $x = 2$.
 $f(1) = \ln 1 - 2 + 1 = -1$. $f(2) = \ln 2 - 2 + 2 = \ln 2 \approx 0.693$. Since $f(1) < 0$ and $f(2) > 0$, the sign change confirms a root lies in $(1, 2)$.
 - 4 Show that $f(x) = x^3 - 2x - 5$ has a root between $x = 2$ and $x = 3$.
 $f(2) = 8 - 4 - 5 = -1$. $f(3) = 27 - 6 - 5 = 16$. Since $f(2) < 0$ and $f(3) > 0$, the sign change confirms a root lies in $(2, 3)$.
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Interval bisection

- 5 Using $f(x) = x^3 - x - 1$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval containing the root.
 $f(1.5) = 3.375 - 1.5 - 1 = 0.875 > 0$. Since $f(1) < 0$ and $f(1.5) > 0$, the root lies in $(1, 1.5)$.
 - 6 Continuing from the previous problem, evaluate $f(1.25)$ and give a narrower interval for the root.
 $f(1.25) = 1.953125 - 1.25 - 1 = -0.296875 < 0$. Since $f(1.25) < 0$ and $f(1.5) > 0$, the root lies in $(1.25, 1.5)$.
 - 7 Using $f(x) = \ln x - 2 + x$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval for the root.
 $f(1.5) = \ln 1.5 - 2 + 1.5 = 0.405465 - 0.5 = -0.094535 < 0$. Since $f(1.5) < 0$ and $f(2) > 0$, the root lies in $(1.5, 2)$.
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Fixed-point iteration

- 8 The equation $x^3 - 4x - 2 = 0$ rearranges to $x = \sqrt[3]{4x + 2}$. Using $x_0 = 2$, find x_1 , x_2 , and x_3 , to four decimal places.
 $x_1 = \sqrt[3]{4(2) + 2} = \sqrt[3]{10} \approx 2.1544$. $x_2 = \sqrt[3]{4(2.1544) + 2} = \sqrt[3]{10.6176} \approx 2.1978$.
 $x_3 = \sqrt[3]{4(2.1978) + 2} = \sqrt[3]{10.7912} \approx 2.2098$ (converging toward the root ≈ 2.214).

- 9 The equation $x^2 - 5x + 2 = 0$ rearranges to $x = 5 - \frac{2}{x}$. Using $x_0 = 4$, find x_1 and x_2 , to four decimal places.
 $x_1 = 5 - \frac{2}{4} = 4.5$. $x_2 = 5 - \frac{2}{4.5} \approx 5 - 0.4444 = 4.5556$.
- 10 The equation $x^3 - 2x - 5 = 0$ rearranges to $x = \sqrt[3]{2x + 5}$. Using $x_0 = 2$, find x_1 , to four decimal places.
 $x_1 = \sqrt[3]{2(2) + 5} = \sqrt[3]{9} \approx 2.0801$.

Convergence and divergence of iterations

- 11 Explain why the iteration $x_{n+1} = x_n^3 - 3$ (i.e. $g(x) = x^3 - 3$), starting from $x_0 = 1.5$, would fail to converge to a root of $x^3 - x - 3 = 0$ near $x = 1.5$.
 $g'(x) = 3x^2$. At $x = 1.5$: $g'(1.5) = 3(2.25) = 6.75$. Since $|g'(1.5)| = 6.75 > 1$, the iteration does not satisfy the convergence condition $|g'(x)| < 1$ near this point, so it diverges rather than converging to the root.
- 12 Using $g(x) = 5 - \frac{2}{x}$ from the earlier iteration, find $g'(x)$ and evaluate $|g'(x)|$ at $x = 4.56$ (near the root), to explain why the iteration converges there.
 $g(x) = 5 - 2x^{-1}$, so $g'(x) = \frac{2}{x^2}$. At $x = 4.56$: $g'(4.56) = \frac{2}{20.7936} \approx 0.0962$. Since $|g'(4.56)| < 1$, the iteration converges near this root.

Rearranging equations for iteration

- 13 Rearrange $x^2 - 3x - 1 = 0$ into the form $x = g(x)$ by making the linear term the subject.
From $3x = x^2 - 1$: $x = \frac{x^2 - 1}{3}$.
- 14 Rearrange $x^3 + x - 1 = 0$ into the form $x = \frac{1}{x^2 + 1}$, and verify algebraically that this is equivalent to the original equation.
From $x^3 + x - 1 = 0$: $x(x^2 + 1) = 1$, so $x = \frac{1}{x^2 + 1}$. Verifying: multiplying both sides by $(x^2 + 1)$ gives $x(x^2 + 1) = 1$, i.e. $x^3 + x = 1$, i.e. $x^3 + x - 1 = 0$, which matches the original equation exactly.

The Newton-Raphson method

- 15 Use one iteration of the Newton-Raphson method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 4x - 2$, to four decimal places.
 $f'(x) = 3x^2 - 4$. $f(2) = 8 - 8 - 2 = -2$. $f'(2) = 12 - 4 = 8$. $x_1 = 2 - \frac{-2}{8} = 2.25$.
- 16 Continuing from $x_1 = 2.25$ in the previous problem, find x_2 , to four decimal places.
 $f(2.25) = 11.390625 - 9 - 2 = 0.390625$. $f'(2.25) = 3(5.0625) - 4 = 11.1875$.
 $x_2 = 2.25 - \frac{0.390625}{11.1875} \approx 2.25 - 0.0349 \approx 2.2151$.

- 17 Use the Newton-Raphson method with $x_0 = 2.5$ to find x_1 for $f(x) = x^2 - 7$ (approximating $\sqrt{7}$), to four decimal places.
 $f'(x) = 2x$. $f(2.5) = 6.25 - 7 = -0.75$. $f'(2.5) = 5$. $x_1 = 2.5 - \frac{-0.75}{5} = 2.5 + 0.15 = 2.65$.
- 18 Use the Newton-Raphson method with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 2x - 5$, to four decimal places.
 $f'(x) = 3x^2 - 2$. $f(2) = 8 - 4 - 5 = -1$. $f'(2) = 12 - 2 = 10$. $x_1 = 2 - \frac{-1}{10} = 2.1$.

Further Newton-Raphson: rapid convergence and failure cases

- 19 Using $x_1 = 2.65$ from the previous problem for $f(x) = x^2 - 7$, perform a second iteration to find x_2 , to four decimal places, and compare with $\sqrt{7} \approx 2.6458$.
 $f(2.65) = 7.0225 - 7 = 0.0225$. $f'(2.65) = 5.3$. $x_2 = 2.65 - \frac{0.0225}{5.3} \approx 2.65 - 0.00425 \approx 2.6458$, matching $\sqrt{7}$ to four decimal places after just two iterations.
- 20 Explain why $x_0 = 1$ is an unsuitable starting point for the Newton-Raphson method applied to $f(x) = x^3 - 3x + 1$.
 $f'(x) = 3x^2 - 3$. At $x_0 = 1$: $f'(1) = 3 - 3 = 0$. Since $f'(1) = 0$, the tangent line at $x_0 = 1$ is horizontal and never meets the x -axis, so the formula $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ involves division by zero — the method fails at this starting point.
- 21 Continuing from $x_1 = 2.1$ for $f(x) = x^3 - 2x - 5$, perform a second Newton-Raphson iteration to find x_2 , to four decimal places.
 $f(2.1) = 9.261 - 4.2 - 5 = 0.061$. $f'(2.1) = 3(4.41) - 2 = 11.23$.
 $x_2 = 2.1 - \frac{0.061}{11.23} \approx 2.1 - 0.00543 \approx 2.0946$.

Error bounds from bisection

- 22 A root of $f(x) = x^3 - 4x - 2$ lies in $(2.21, 2.22)$, since $f(2.21) \approx -0.046$ and $f(2.22) \approx 0.061$. State the midpoint estimate and the maximum possible error.
Midpoint: $\frac{2.21 + 2.22}{2} = 2.215$. Maximum possible error: half the interval width, $\frac{2.22 - 2.21}{2} = 0.005$.
- 23 Given that the root of $f(x) = x^3 - 4x - 2$ is trapped in the interval $(2.2141, 2.2144)$ after further bisection, state the root correct to three decimal places, justifying your answer.
Every value in the interval $(2.2141, 2.2144)$ rounds to 2.214 at three decimal places (since both endpoints lie strictly between 2.2135 and 2.2145). The root is therefore 2.214 to three decimal places.
- 24 A root of $f(x) = x^3 - 2x - 5$ lies in $(2.09, 2.10)$, since $f(2.09) \approx -0.051$ and $f(2.10) \approx 0.061$. State the midpoint estimate and the maximum possible error.
Midpoint: $\frac{2.09 + 2.10}{2} = 2.095$. Maximum possible error: $\frac{2.10 - 2.09}{2} = 0.005$ (consistent with the Newton-Raphson estimate $x_2 \approx 2.0946$ found earlier).

A well-designed word problem: designing a support strut

- 25** This question has four parts. An engineer models the cross-sectional profile of a support strut by $f(x) = x^3 - 6x^2 + 9x - 2$ (metres) and needs to find where it crosses the x -axis. (a) Show by a change of sign that a root lies between $x = 0$ and $x = 1$. (b) Use the Newton-Raphson method with $x_0 = 0.5$ to find x_1 , to four decimal places. (c) Perform a second iteration to find x_2 , to four decimal places. (d) Perform a third iteration to find x_3 , and state the root to three decimal places.

$f'(x) = 3x^2 - 12x + 9$. (a) $f(0) = -2$ and $f(1) = 1 - 6 + 9 - 2 = 2$; the sign change confirms a root in $(0, 1)$. (b) $f(0.5) = 0.125 - 1.5 + 4.5 - 2 = 1.125$; $f'(0.5) = 0.75 - 6 + 9 = 3.75$;
 $x_1 = 0.5 - \frac{1.125}{3.75} = 0.2$. (c) $f(0.2) = 0.008 - 0.24 + 1.8 - 2 = -0.432$;
 $f'(0.2) = 0.12 - 2.4 + 9 = 6.72$; $x_2 = 0.2 - \frac{-0.432}{6.72} \approx 0.2643$. (d) $f(0.2643) \approx -0.02196$;
 $f'(0.2643) \approx 6.0380$; $x_3 = 0.2643 - \frac{-0.02196}{6.0380} \approx 0.2679$. The root is 0.268 to three decimal places.